MAE 303, Project Title:

Development of an Autonomous Line-following Soccer Robot

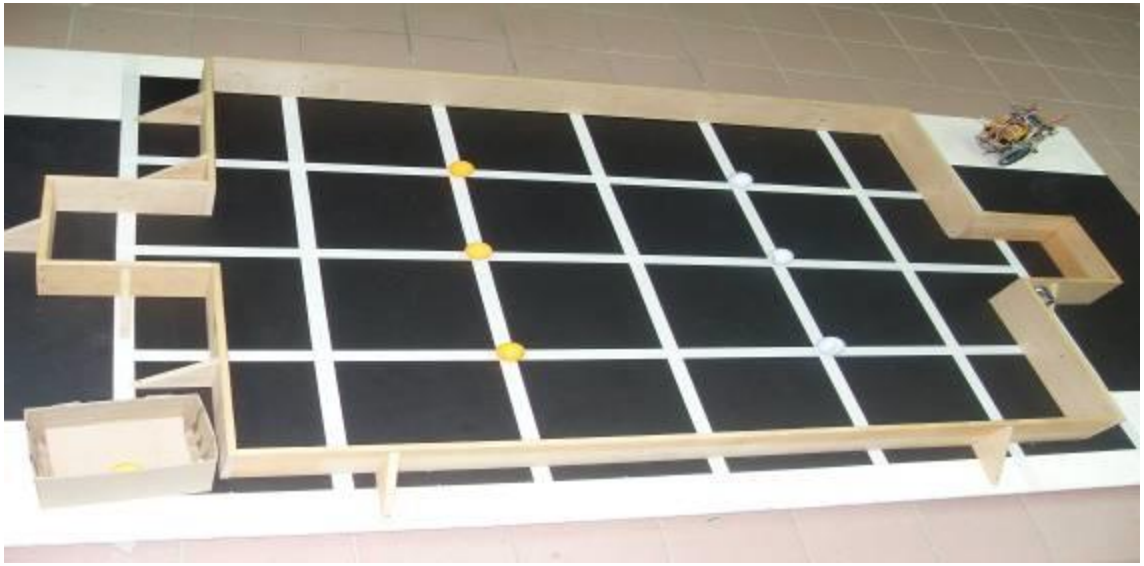
INTRODUCTION

- One of the applications of Modern Control Engineering is to build an autonomous robot.
- It requires one to integrate the field of mechanical, programming, and control systems to build a functional robot.
- In this subject, each group of students is assigned to build a soccer robot for internal competition.
- Students are required to build a soccer robot based on their knowledge and skills with creativity.

OBJECTIVES

- To build a soccer robot that will be able to navigate and push its balls toward the opponent's goal
- To build a smart/intelligent robot that can avoid obstacles (i.e. wall or opponent's balls and robot)
- To develop tactics and strategies for opponents

GAME FIELD LAYOUT



- Field size: 150 cm x 100 cm
- Black field with white lines
- Ball types: ping-pong balls, golf balls
- The robot should not exceed a length of 20 cm.

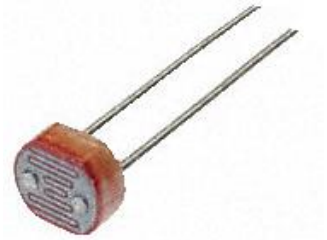
RULES OF THE GAME

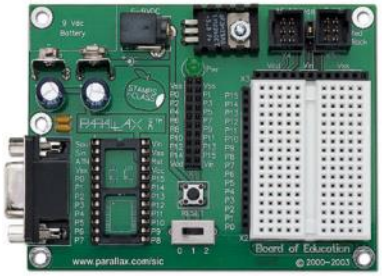
- Each game lasts for 3 minutes
- Push your balls to the opponent's goal
- Both teams will start at the same time
- There is no limit to restart
- If you push the opponent's balls to their goals, consider a goal for them
- Only 1 robot can be used for 1 session
- A robot which does not move at all, to be considered as a loser.

DESIGN OVERVIEW

Line Following:

Light Dependent Resistor (LDR)





Microcontroller:

Any type

Obstacle Detection:

Sharp Analog Distance Sensor



Actuators:

Servos/DC Motors



